

GROWTH INTERN ASSIGNMENT

Jar — Growth Intern Assignment

Sales analysis · App exploration · Product exploration

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Code: <https://github.com/VT69/Jar---Assignment>

Analysis in Python · all figures and numbers regenerated by `python main.py`

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Executive summary

Categories. Electronics leads on sales (₹1,65,267, 38.3%) and profit per order (₹51.44). Clothing earns the most profit (₹11,163) at the best margin (8.03%). Furniture trails on every metric (1.81% margin).

Profit leaks. Two sub-categories lose money: Tables (-₹4,011) and Electronic Games (-₹1,236). In 16 of 17 sub-categories, loss-making lines realise lower unit prices than profitable ones (median 0.55×) at the same quantity.

Furniture targets. The target ramps smoothly (no MoM move above 1.89%), but demand is back-loaded. Furniture missed target 7 months in a row, then beat it in 4 of the last 5 (95.7% for the year). Seasonal phasing cuts monthly error from 54.4% to 27.3% (19.5% in-sample).

Regions. Madhya Pradesh and Maharashtra hold 38.2% of orders. Mumbai's margin is 2.65%, against 13.56% in Pune. 4 states lose money (-₹3,642), while Uttar Pradesh, Delhi, West Bengal and Kerala earn 44.2% of profit on 16.8% of sales.

App (Q2). The most serious issue is a silent failure: Nominee Details does nothing when tapped from Profile, though it works from a home-screen pop-up. Reward screens such as a spin wheel also interrupt users mid-task. Onboarding, AutoPay saving and buying and selling gold were smooth.

Product (Q3). Digital gold sits outside SEBI's oversight industry-wide (advisory, November 2025). Jar can lead the category by setting the disclosure standard and moving its automation onto regulated rails. The first moves are connecting existing goals to Nek redemption and a SEBI-regulated gold ETF/FoF, followed by insured deposits, sachet insurance and partner-led gold credit.

1. Data and method

1.1 Join validation

Order Details (one row per line item) is **left-joined** to List of Orders on Order ID with pandas `validate="many_to_one"` and an indicator column. Order Details carries the grain of every financial metric. The many-to-one check proves Order ID is unique on the header side, so the join cannot duplicate lines. A hard assertion fails the run on any orphan, instead of letting an inner join drop it silently.

Check	Result
List of Orders rows / unique Order IDs	500 / 500
Order Details rows / unique Order IDs	1,500 / 500
Duplicate Order IDs in List of Orders	0
IDs only in List of Orders / only in Order Details	0 / 0
Rows after merge	1,500 (equals line items)

Orders have 1 to 12 lines, and 207 span more than one category. That is why "per order" and "per line" averages differ.

1.2 Date repairs

Spreadsheet auto-conversion corrupted both date fields. Both were repaired in code, with assertions; the raw files are untouched.

- **Order Date.** 307 cells are `dd-mm-yyyy` text. The other 193 were read month-first, so day and month were swapped back. Repaired dates rise monotonically with the sequential Order ID and span 01 Apr 2018 to

31 Mar 2019.

- **Sales Target month.** "Apr-18" had been stored as *18 April of the current year*, so year = 2000 + stored day. The months run April 2018 to March 2019, matching the order period exactly.

1.3 Definitions

- **Sales** = sum of `Amount`. **Margin** = $\text{sum(Profit)} \div \text{sum(Amount)}$. **Order count** = distinct Order IDs.
- **Average profit per order (primary)** = $\text{group profit} \div \text{distinct orders containing the group}$. **Per line item (secondary)** = mean line profit. Both rank the categories identically; per-order values are 1.3×–2.4× higher.
- **Price realisation.** The data has no discount field, so discounting can only be tested indirectly: realised unit price ($\text{Amount} \div \text{Quantity}$) on loss-making lines is compared with profitable lines in the same sub-category.

1.4 Sample size

The dataset is small: 500 orders and 1,500 lines over one year. Several cuts rest on few observations: Tables has 17 lines, Tamil Nadu 8 orders, and Furniture sells 10–32 orders a month. A handful of large lines can move a sub-category, city or month. Category-level findings are robust. Single-city and single-month results are directional and should be confirmed with more data before acting.

2. Q1 Part 1 — Sales and profitability

2.1 Category scorecard

Electronics leads on sales and profit per order; Furniture trails on every metric

Category totals across all 500 orders, Apr 2018 - Mar 2019

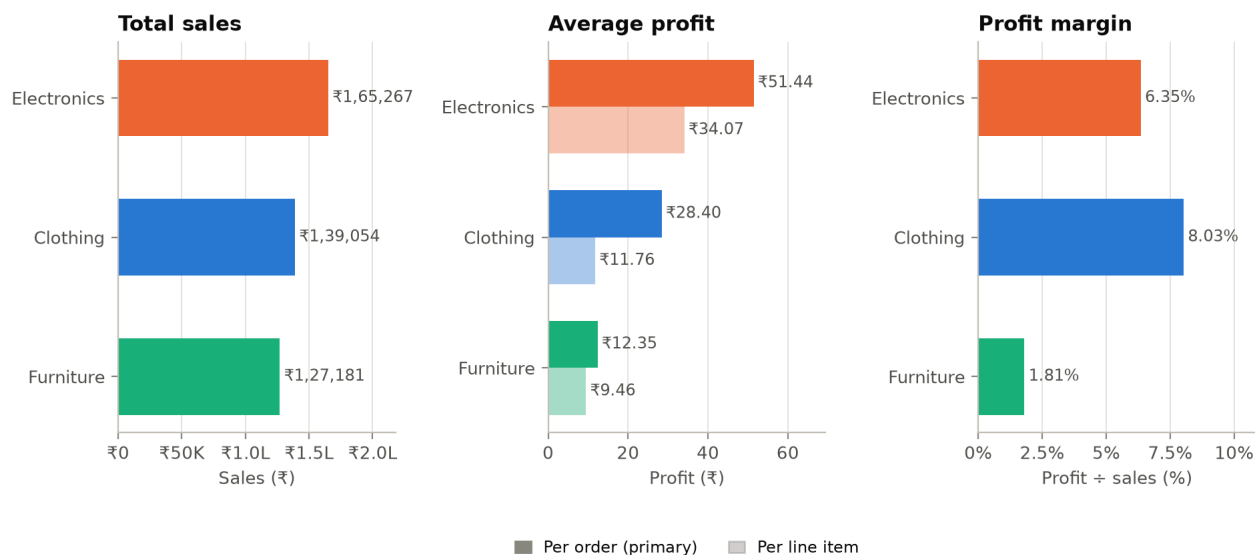


Figure 1. Category totals: sales, average profit (per order and per line) and margin.

Category	Sales	Share of sales	Profit	Share of profit	Profit / order	Profit / line	Margin
Electronics	₹1,65,267	38.3%	₹10,494	43.8%	₹51.44	₹34.07	6.35%
Clothing	₹1,39,054	32.2%	₹11,163	46.6%	₹28.40	₹11.76	8.03%
Furniture	₹1,27,181	29.5%	₹2,298	9.6%	₹12.35	₹9.46	1.81%
Total	₹4,31,502	100%	₹23,955	100%			5.55%

Verdict.

- **Electronics leads on scale and order economics:** #1 on sales and on profit per order.
- **Clothing leads on efficiency:** #1 on total profit and on margin. Its per-line profit (₹11.76) understates it, because its lines are small (unit price ₹39.55). It also appears in 393 of 500 orders, the most of any category.
- **Furniture underperforms:** last on all four metrics.

2.2 Why the categories differ

1. One or two sub-categories hold each category back.

Two sub-categories lose money: Tables (-₹4,011) and Electronic Games (-₹1,236)

Total profit by sub-category; bars sorted within each category

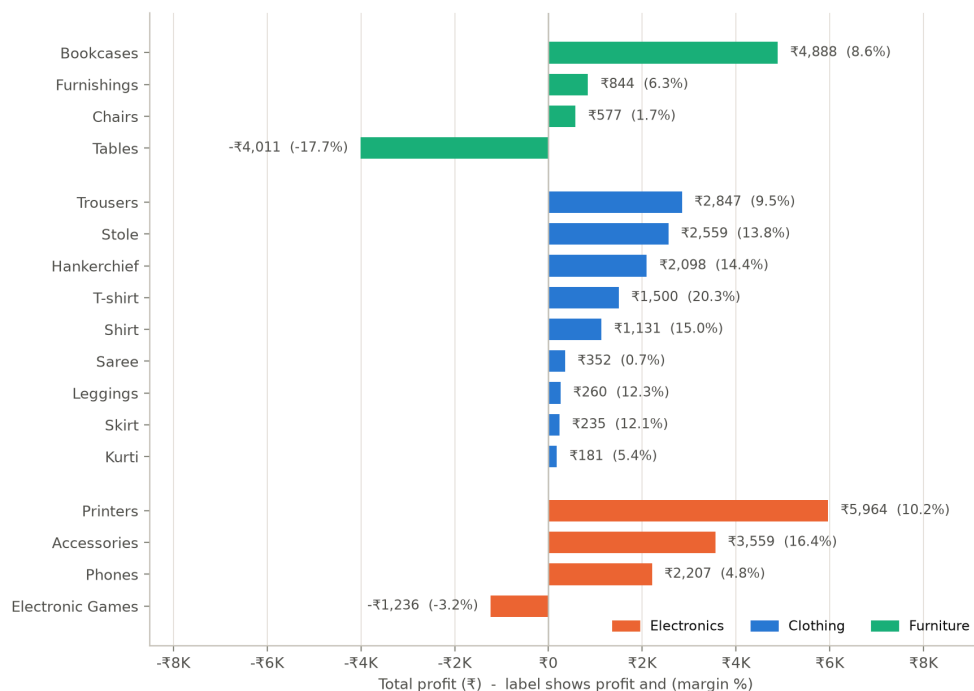


Figure 2. Profit and margin by sub-category.

- **Furniture.** Tables lose -₹4,011 on ₹22,614 (-17.7% margin; 64.7% of lines in loss). Chairs barely break even (1.7%). Bookcases earn ₹4,888, which is 213% of Furniture's net profit. Without Tables, the margin would be 6.03%, close to Electronics (6.35%).
- **Electronics.** Electronic Games lose -₹1,236 (-3.2%), and Phones earn 4.8% with 49.4% of lines in loss. Printers (56.8% of category profit) and Accessories (16.4% margin) carry the category. Without Games, the margin would be 9.30%.
- **Clothing.** Sarees are 38.5% of sales at a 0.66% margin. The rest of Clothing earns 12.6%.

2. Losses are concentrated, most severely in Furniture.

Furniture's loss-making lines erase 87% of what its profitable lines earn

Line-level gains and losses by category

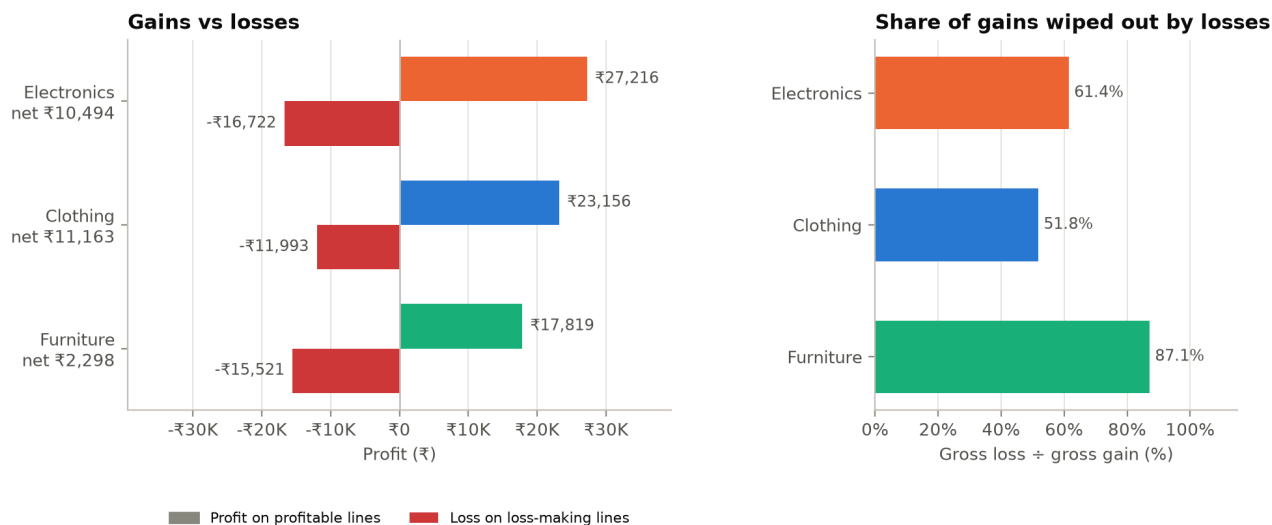


Figure 3. Gross gains from profitable lines vs gross losses from loss-making lines.

Category	Lines in loss	Loss ÷ gain	Worst 10 lines' share of losses	Lines losing more than their revenue
Electronics	39.6%	61.4%	41.5%	3
Clothing	29.4%	51.8%	28.7%	1
Furniture	42.0%	87.1%	57.6%	10

Furniture's loss-making lines give back ₹15,521 of the ₹17,819 its profitable lines earn. Its worst 10 lines account for 57.6% of those losses.

3. Loss-making lines realise lower prices, not smaller quantities.

In 16 of 17 sub-categories, loss lines sell at a lower unit price (median 0.55x)

Price-realisation check: no discount field exists, so realised unit price (Amount ÷ Quantity) is compared

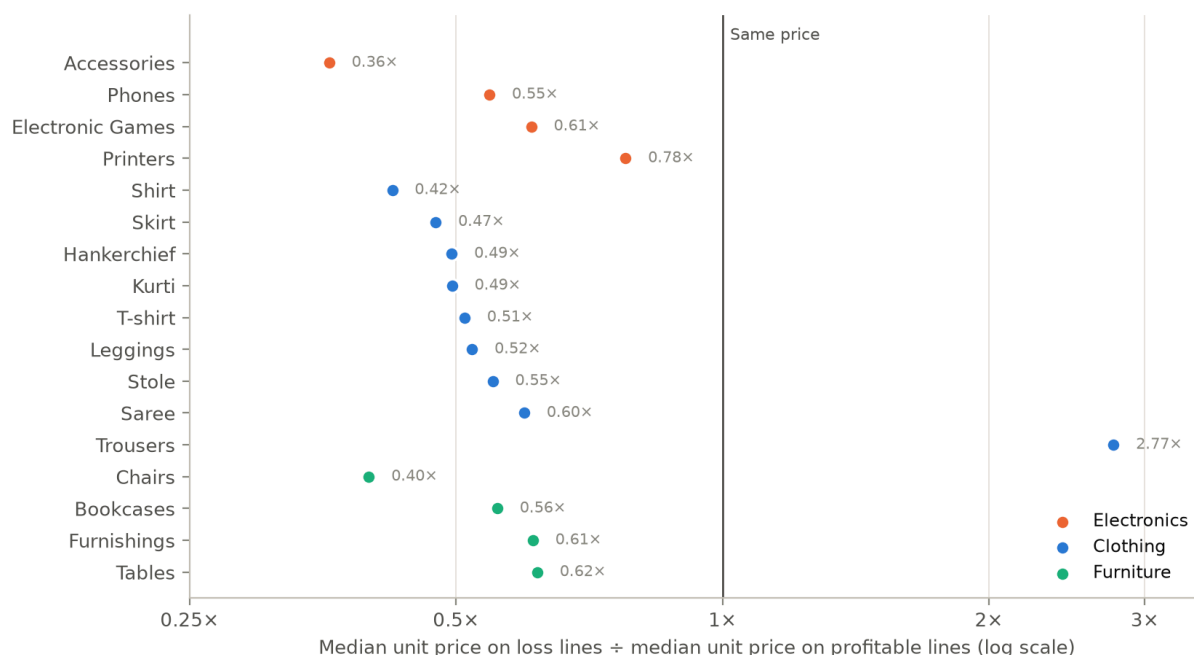


Figure 4. Median unit price on loss-making vs profitable lines within each sub-category.

In 16 of 17 sub-categories, loss-making lines sell at a lower median unit price than profitable ones (median ratio 0.55x; Chairs 0.40x). The exception is Trousers (2.77x), where losses sit on high-ticket lines, which suggests a cost problem. Median quantity is the same on both (3 units), so the losses track **low realised prices**, not volume.

The data cannot say *why* prices are low. Discounting (the same item sold for less) is one hypothesis. **Product mix** is the alternative: cheaper variants within a sub-category may carry thin or negative margins. Separating the two needs SKU-level list prices.

4. Furniture keeps little profit per unit. Furniture and Electronics sell at similar unit prices (₹134.58 vs ₹143.21). Furniture keeps ₹2.43 per unit against ₹9.09 for Electronics, so a small price concession tips a Furniture line into loss.

Actions.

1. Pull SKU-level list prices to test whether losses are discount-driven or mix-driven.
2. If discount-driven, set price floors on Tables and Chairs and a floor on Saree realised prices. If mix-driven, re-cost or prune the loss-making variants instead.
3. Re-price or delist Electronic Games.

3. Q1 Part 2 — Target achievement (Furniture)

3.1 Month-over-month change in target

The Furniture target ramps smoothly: every MoM change is between 0.87% and 1.89%

Red = flagged by the z-score rule ($|\text{MoM \%} - \text{mean}| > 1.5 \text{ SD}$)

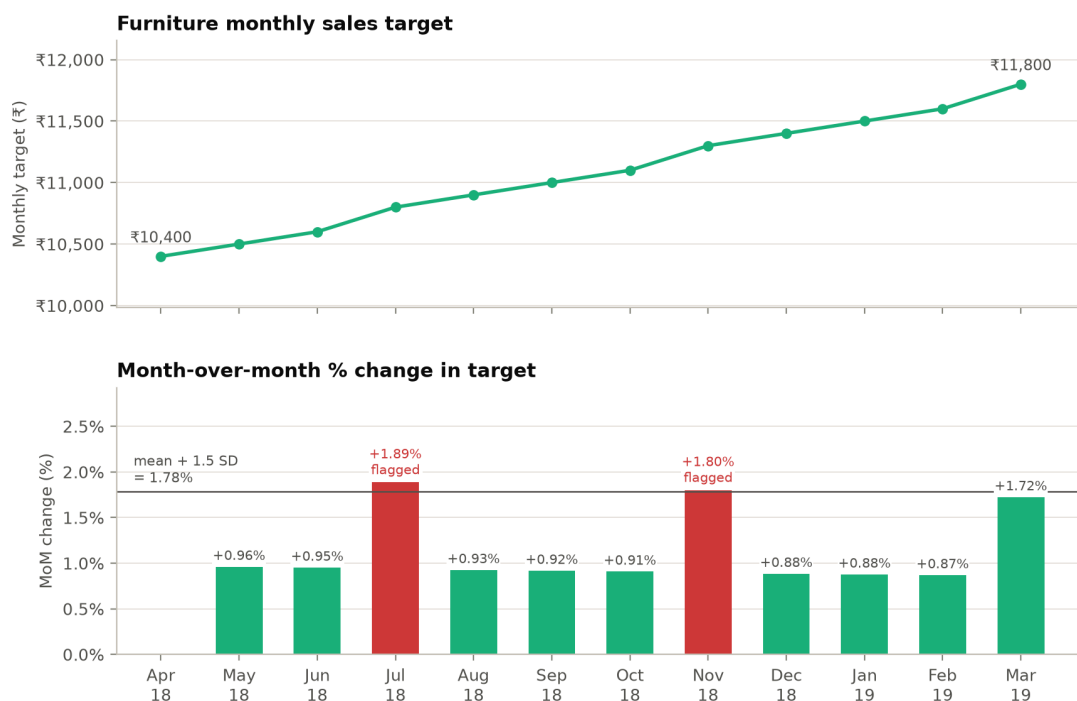


Figure 5. Furniture monthly target and its month-over-month % change.

Fluctuation rule. A month is flagged when its MoM % lies more than 1.5 standard deviations from the series mean (z-score; threshold 1.78%). As a materiality cross-check, it is also flagged if $|\text{MoM}| \geq 5\%$.

Month	Target	Change (₹)	MoM %	z-score	Flag (z-rule)
Apr 2018	₹10,400	—	—	—	
May 2018	₹10,500	+₹100	+0.96%	-0.46	
Jun 2018	₹10,600	+₹100	+0.95%	-0.48	
Jul 2018	₹10,800	+₹200	+1.89%	+1.74	Flagged
Aug 2018	₹10,900	+₹100	+0.93%	-0.55	
Sep 2018	₹11,000	+₹100	+0.92%	-0.57	
Oct 2018	₹11,100	+₹100	+0.91%	-0.59	
Nov 2018	₹11,300	+₹200	+1.80%	+1.54	Flagged
Dec 2018	₹11,400	+₹100	+0.88%	-0.65	
Jan 2019	₹11,500	+₹100	+0.88%	-0.66	
Feb 2019	₹11,600	+₹100	+0.87%	-0.68	
Mar 2019	₹11,800	+₹200	+1.72%	+1.36	

The target rises 13.5% across the year, in steps of ₹100, with ₹200 steps in Jul 2018, Nov 2018 and Mar 2019. The z-rule flags **Jul 2018 and Nov 2018**, with Mar 2019 just under the bar. The materiality rule flags **0 months**. The target's fluctuations are statistically visible but commercially trivial. The problem is a target *too smooth* for a volatile business.

3.2 Target vs actual

Flat targets miss a back-loaded year: 7 straight misses, then 4 of the last 5 months above target

Actual = sum of Furniture line Amounts per repaired order month

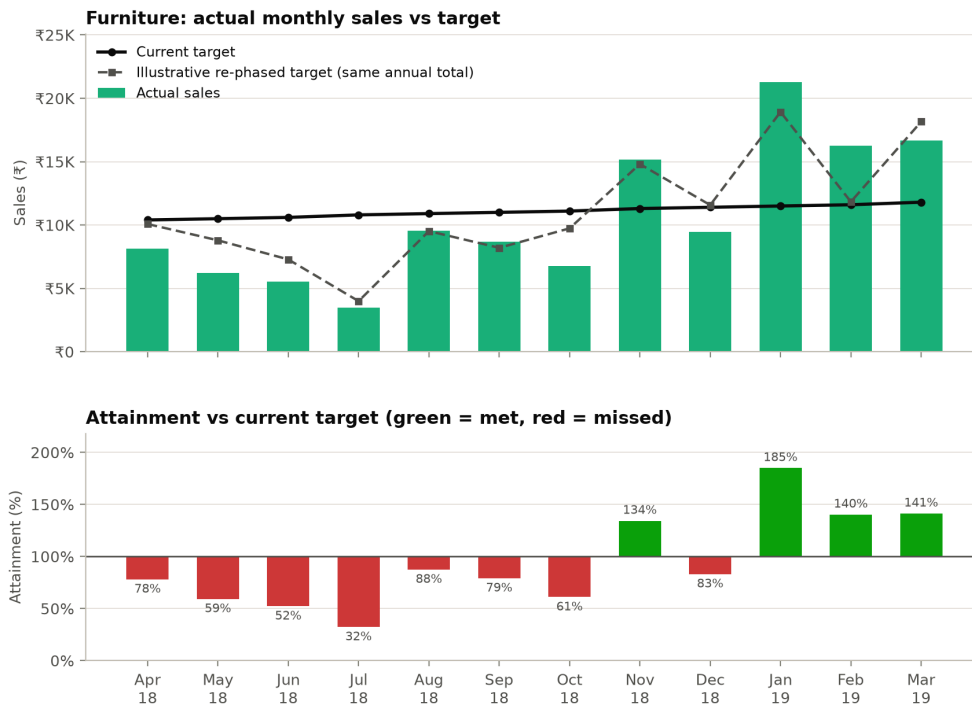


Figure 6. Actual Furniture sales vs current and illustrative re-phased target, and monthly attainment.

- **Annually about right:** ₹1,27,181 achieved against ₹1,32,900 (95.7%).
- **Monthly, phased wrong.** Target was met in only 4 of 12 months. Furniture missed for 7 straight months to Oct 2018 (64.2% attainment, -₹26,936). It then beat target in 4 of the last 5 (136.8%, +₹21,217).
- **H1 over-weighted.** April–September carries 48.3% of the target but only 32.7% of actual sales.
- **Actuals are 12.5× more volatile than the target.** The coefficient of variation is 51.3% for actuals vs 4.1% for the target.
- **Revenue-only targets hide losses.** Furniture lost money in 6 months (-₹9,316 in total).

3.3 Aligning targets with performance

1. **Phase by seasonality.** Spread the same ₹1,32,900 across months by each month's share of company-wide sales. The monthly target then ranges from ₹3,993 (Jul 2018) to ₹18,923 (Jan 2019). MAPE falls from 54.4% to 19.5%, and months within $\pm 15\%$ of target rise from 1 to 6.
 - 19.5% is optimistic: company sales include Furniture itself (29.5% of the total), and seasonality comes from the same year. With Clothing and Electronics alone, which excludes Furniture entirely, MAPE is 27.3%, the better estimate. In practice, phase on prior-year or rolling three-year shares.
2. **Add a margin guardrail** (e.g. margin $\geq 0\%$ each month), and tie price approvals on Tables and Chairs to it.
3. **Use a $\pm 15\%$ band and re-forecast quarterly.** With an actual CV of 51%, a point target will be missed almost every month.

4. **Re-balance across categories.** Electronics hit 128.1% of target and Clothing 79.9%. Shift ambition toward demonstrated run-rates.

4. Q1 Part 3 — Regional performance

4.1 Top 5 states by order count

Two states hold 38.2% of orders; Punjab makes the top 5 yet loses money

Top 5 states by distinct Order ID count. Red = negative

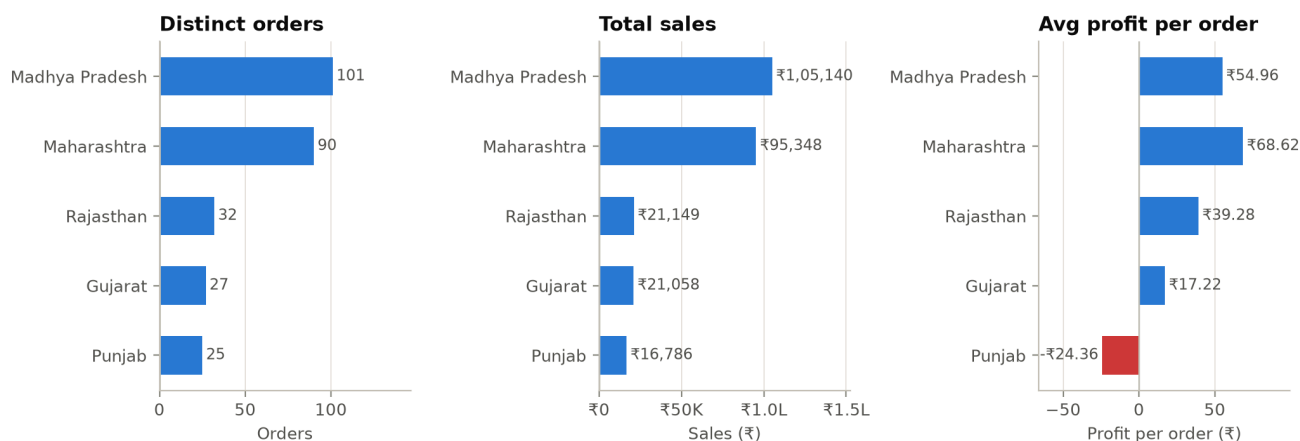


Figure 7. Top 5 states by distinct orders: order count, total sales and average profit per order.

Rank	State	Distinct orders	Total sales	Total profit	Avg profit / order	Avg profit / line	Margin
1	Madhya Pradesh	101	₹1,05,140	₹5,551	₹54.96	₹16.33	5.28%
2	Maharashtra	90	₹95,348	₹6,176	₹68.62	₹21.30	6.48%
3	Rajasthan	32	₹21,149	₹1,257	₹39.28	₹16.99	5.94%
4	Gujarat	27	₹21,058	₹465	₹17.22	₹5.34	2.21%
5	Punjab	25	₹16,786	-₹609	-₹24.36	-₹10.15	-3.63%

Average profit is per order (state profit ÷ distinct orders); per-line is shown for reference. There is no tie at the cut-off: the next states (Delhi, Uttar Pradesh and West Bengal) have 22 orders each. The top 5 hold 55.0% of orders and 60.1% of sales, but only 53.6% of profit.

4.2 Regional disparities

4 states lose money; 4 smaller states earn well above their sales share

Blue = state captures more profit than its sales share; red = less

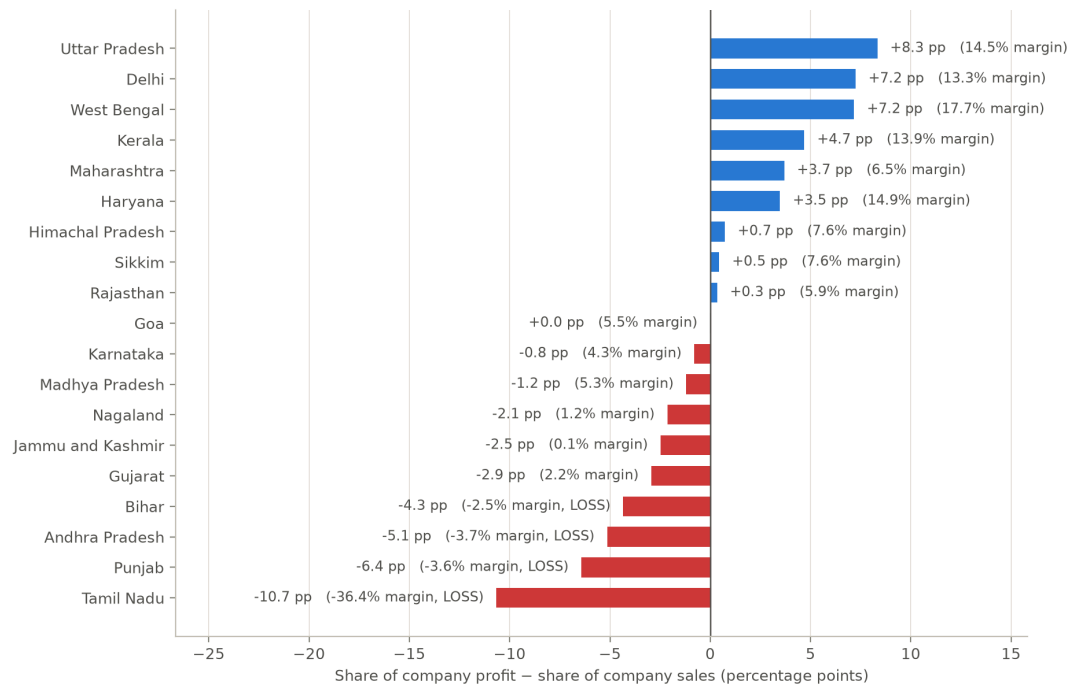


Figure 8. Each state's share of company profit minus its share of company sales.

- **Sales-to-profit mismatch.** Madhya Pradesh has 24.4% of sales but 23.2% of profit. Uttar Pradesh, Delhi, West Bengal and Kerala earn 44.2% of profit on 16.8% of sales, a 14.6% margin against 5.55% for the company.
- **Loss-making states, each driven by one category.**
 - Tamil Nadu: -₹2,216, from Furniture.
 - Punjab: -₹609, from Electronics (-₹2,132).
 - Andhra Pradesh: -₹496, from Furniture.
 - Bihar: -₹321, from Electronics.
- **City-level splits.** In Gujarat, Punjab and Rajasthan, one city makes money and the other loses it:

State	Loss-making city	Sales	Profit	Margin	Profitable city	Margin
Gujarat	Ahmedabad	₹14,230	-₹880	-6.2%	Surat	19.7%
Punjab	Chandigarh	₹12,279	-₹1,153	-9.4%	Amritsar	12.1%
Rajasthan	Jaipur	₹10,076	-₹753	-7.5%	Udaipur	18.2%

Mumbai and Indore carry the volume at thin margins; Chennai is the deepest loss

Each bubble is a city (size = distinct orders). Labels show priority cities and their margin

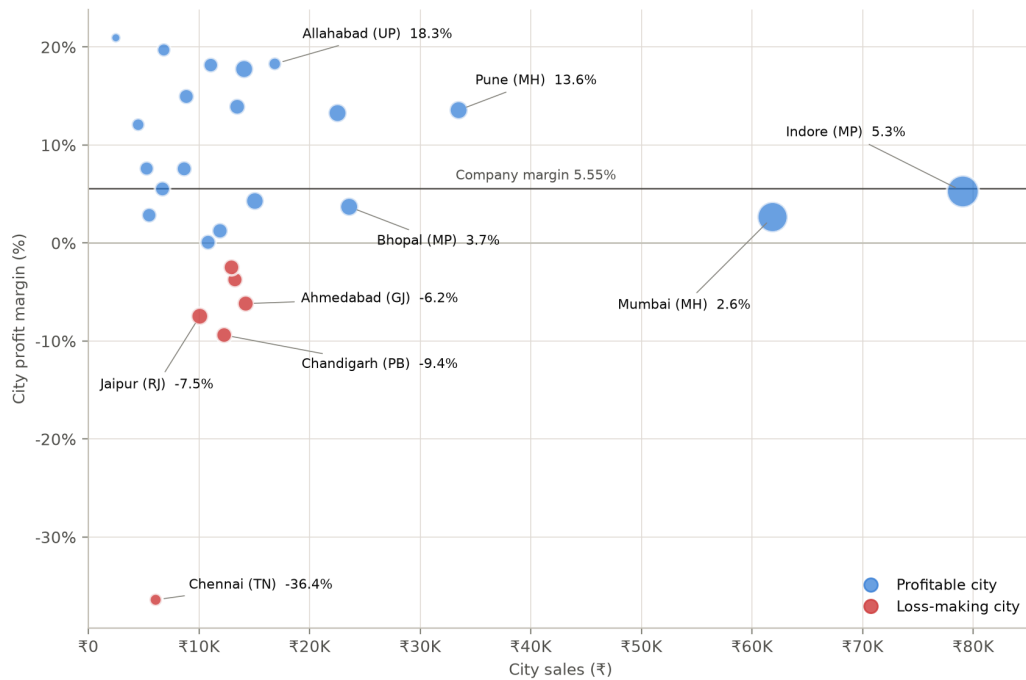


Figure 9. City sales vs margin; bubble size is distinct orders.

4.3 Where to focus

Priority	Where	Why (number)	Lever
1. Margin repair at scale	Mumbai	₹61,867 at 2.65%; 41.1% of lines in loss vs 28.9% in Pune	Pune's margin would add about ₹6,750; the company average, ₹1,798
2. Investigate	Chennai (Tamil Nadu)	-₹2,216 on ₹6,087 (-36.4%) from only 8 orders	Find what drives the Furniture loss (-₹2,487): pricing, freight or one-off returns. Act once the cause is confirmed
3. Fix loss cities in the top 5	Ahmedabad, Chandigarh and Jaipur	-₹2,786 combined; sister cities earn double-digit margins	Copy the sister city's mix and pricing; for Punjab, start with Electronics
4. Defend the volume base	Indore, Bhopal	Indore is 75.2% of Madhya Pradesh sales; Bhopal earns 3.7%	Hold Indore near the company margin; lift Bhopal
5. Grow where profit is easy	Allahabad, Delhi, Kolkata, Thiruvananthapuram	14.6% combined margin on only 82 orders	Spend acquisition budget here first

Data note: 3 orders list city "Delhi" under Madhya Pradesh; they are kept as recorded and change no ranking. Chandigarh appears under both Punjab (16 orders) and Haryana (14).

5. Q2 — App exploration

Test setup. Hands-on session on a Vivo T2x 5G (6 GB RAM) over Jio 5G. Jar blocks screenshots, a standard security measure in fintech apps, so observations are described rather than shown.

5.1 Five things that work well

#	What I observed	Why it works (UX principle)
1	Onboarding was simple.	<i>Low onboarding friction.</i> For first-time savers, every step removed before the first saving raises the share who get there.
2	Daily savings on AutoPay worked smoothly, and a goals feature exists.	<i>Habit loop and default effect.</i> A one-time mandate replaces a daily decision, so saving runs without willpower. Goals give the habit a purpose.
3	Buying and selling gold was smooth.	<i>Reversibility lowers perceived risk.</i> Users commit money more readily when getting it back is easy.
4	The payment screen carries a trust line: "100% safe and secure payments".	<i>Trust signals in fintech.</i> Reassurance at the moment of payment is well placed. But it addresses payment security, not where the gold is held. Showing the vault partner (Brink's), trustee (Vistra) and insurer (ICICI Lombard) here would answer the question users actually have. The website names all three, but the insurer appears only in its body text; its "100% Insured" badge does not name it.
5	Engagement is strong, with multiple reward mechanics.	<i>Variable rewards.</i> Rewards give users a reason to open the app between savings and reinforce the daily habit. Their cost is covered in 5.2 #2.

5.2 Five areas to improve

#	What I observed	Why it matters (UX principle) and what to change
1	Nominee Details is unresponsive from the Profile section. Tapping it does nothing, with no error or feedback. The same flow works from the home-screen pop-up, so one entry point is broken. This is the most serious finding.	<i>Visibility of system status; error recovery</i> (Nielsen). A silent failure leaves users unsure whether anything happened. Here it blocks a basic safeguard for a financial asset, naming who inherits the gold, at the place users expect to find it. Fix the Profile entry point, and give every action that can fail an explicit error state.
2	Reward screens interrupt intent. While I was trying to do something else, a spin wheel appeared first. The multiple reward surfaces also make the app feel cluttered.	<i>Task completion vs interruption.</i> Engagement mechanics are competing with what the user came to do. Show rewards after a task is done, not before it, and consolidate them into fewer surfaces.
3	Several screens load slowly, despite a capable device on 5G.	<i>Perceived performance.</i> In a money app, slow screens read as unreliability. A capable device on 5G rules out the obvious causes on the user's side, which points to the app. Prioritise the saving and payment flows.
4	Visual polish. Several graphics are blurry, and the loader looks unrefined.	<i>Aesthetic-usability effect.</i> Users read polish as care. In a product that holds their money, rough visuals quietly erode trust.
5	No global search across features.	<i>Information scent and findability.</i> As the feature set grows (savings, goals, rewards, Nek), features get harder to find. Search would also reduce reliance on home-screen pop-ups to surface them.

6. Q3 — Product exploration

6.1 Starting point: Jar's strengths and the category's open gap

Jar's moat is behavioural. It has (i) **automation**: UPI AutoPay mandates across more than 35 million registered users; (ii) **micro-ticket access** at ₹10; (iii) **UX simplicity**, in nine languages, for a base that is about 60% tier-2/3; and (iv) **established credibility with first-time savers**, backed by a vault partner (Brink's) and trustee (Vistra) shown in its website's "Secured by" badge, and an insurer (ICICI Lombard) named in the website's body text.

The category has an open gap that every platform shares. SEBI's advisory of 8 November 2025 noted that digital gold is neither a security nor a regulated commodity derivative, so it sits outside SEBI's oversight. Industry voices have since called for standardised disclosure of storage, fees, redemption and insurance. Jar is best placed to lead here. It already names its custody, insurance and trustee partners on its website, and it has the scale to set a disclosure standard others follow. Leading on transparency in digital gold, while extending its automation onto regulated rails, both deepens Jar's credibility and opens new revenue.

Start with Nek goal redemption and SEBI-regulated gold; gate credit behind compliance

Analyst-judgement scores (1-5). Priority = impact ÷ effort

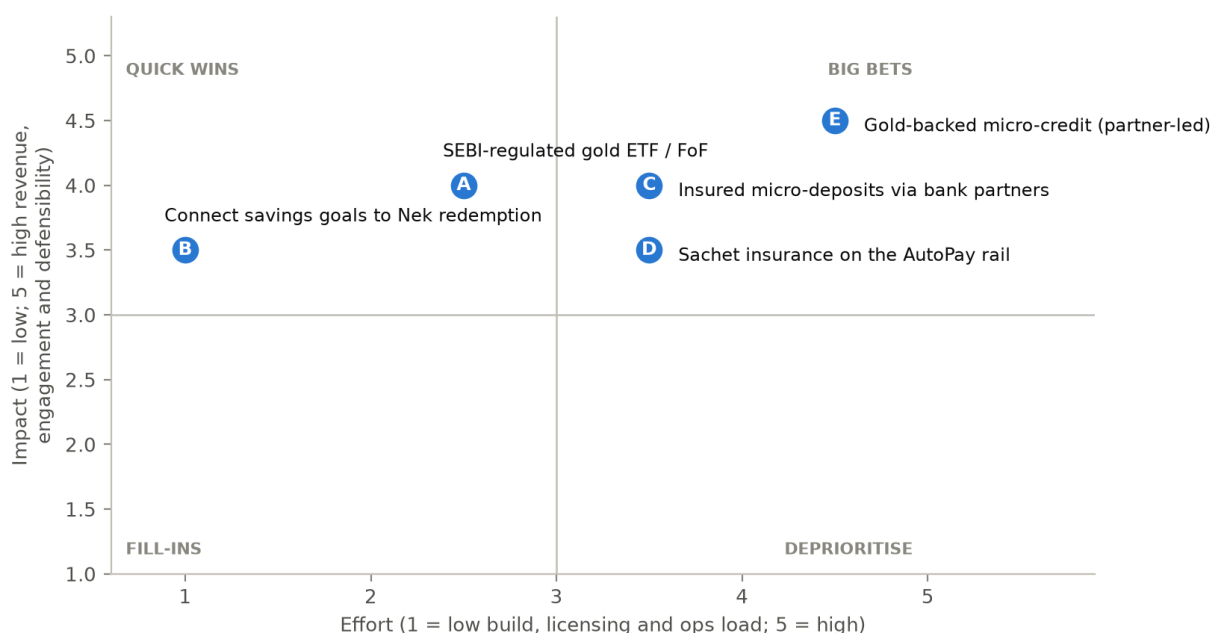


Figure 10. Effort vs impact for the five opportunities (analyst-judgement scores, 1–5).

Rank	Opportunity	Impact	Effort	Quadrant	Horizon
1	B. Connect savings goals to Nek redemption	3.5	1.0	Quick win / do now	Now (0-3 months)
2	A. SEBI-regulated gold ETF / FoF	4.0	2.5	Quick win / do now	Now (0-6 months)
3	C. Insured micro-deposits via bank partners	4.0	3.5	Big bet / sequence	Next (6-12 months)
4	D. Sachet insurance on the AutoPay rail	3.5	3.5	Big bet / sequence	Next (9-15 months)
5	E. Gold-backed micro-credit (partner-led)	4.5	4.5	Big bet / sequence	Later (gated, 12+ months)

6.2 Opportunity cards

A. SEBI-regulated gold ETF / fund-of-funds — the defensive priority

- **What it is.** A SEBI-regulated gold ETF or gold fund-of-funds, held in the user's own name. This is distinct from Jar's existing digital-gold SIP and daily saving, which buy unregulated digital gold.
- **User problem.** Savers who want gold *with* investor protection don't know how to buy an ETF.
- **Why Jar wins.** It already holds the AutoPay mandate and the gold-saving intent. A "Gold+ (SEBI-regulated)" option moves an existing habit rather than acquiring a new customer. Low-ticket mutual-fund SIPs (₹100–500 at many AMCs) fit Jar's micro-ticket DNA.
- **Integration path.** Become a mutual-fund distributor, or partner with a registered platform → add a second "jar" on the home card → route new savings there, with a cost comparison shown.
- **Monetisation.** Distribution trail on regular plans, or a platform fee on direct plans. Lower per-rupee take than the digital-gold spread, but recurring and defensible.
- **Sizing (back-of-envelope).** 35 million is *registered* users, and Jar does not publish its active base, so assume 20% are active savers (70 lakh).
 - *New revenue:* if 10% of active savers (7 lakh) invest ₹500 a month in it, that is about ₹420 crore of regulated inflow in year one. At an indicative 0.3% annual trail, the book earns a year-end run-rate of about ₹1.26 crore a year.
 - *Retention:* FY2024 operating revenue of about ₹208 crore works out to roughly ₹297 per active saver a year. If a regulated option cuts annual churn by 2 percentage points, it keeps 1.4 lakh savers and about ₹4.16 crore a year.
 - *Total:* about ₹5.42 crore a year, or 2.6% of FY2024 operating revenue.
- **Why impact is 4.0.** The quantified value is material but not the largest on the list. The rest is defensibility: keeping the gold habit, and its AutoPay mandate, inside Jar for savers who want investor protection. That is real but hard to size.
- **Key risk.** Cannibalising the digital-gold spread; distributor conduct rules (suitability, commission disclosure).
- **Success metric.** Share of active savers holding the SEBI-regulated product within 6 months; regulated share of gold AUM.

B. Connect savings goals to Nek redemption — the quick win

- **User problem.** Households save for jewellery around weddings, Akshaya Tritiya and Dhanteras, often through jeweller 11-month schemes that lock money to one store.
- **Why Jar wins.** Jar already has a goals feature and its own jewellery brand, Nek. Connecting them turns a completed goal into a purchase the user has been working toward.
- **Integration path.** Goal completion → one-tap Nek purchase, with a making-charge discount for completed goals. Both pieces exist, so the build is the connection, not a new product.
- **Monetisation.** Nek margin and making charges. CAC is lower because the buyer is pre-funded and pre-committed.
- **Sizing (back-of-envelope).** On the same assumed active base (70 lakh), say 5% of active savers (3.5 lakh) save toward a jewellery goal averaging ₹12,000, and 30% complete and redeem at Nek. That gives about 1.1 lakh orders and ₹126 crore of GMV, roughly 1.3× Nek's reported annual revenue of over ₹100 crore.
 - *Margin:* at an assumed 10% gross margin on jewellery, that is about ₹12.6 crore.
- **Key risk.** Must be framed as accumulation of the user's own gold, not a deposit scheme. Fulfilment quality at Nek.
- **Success metric.** Share of completed goals redeemed at Nek; 12-month retention of goal vs non-goal savers.

C. Insured micro-deposits via bank partners

- **User problem.** First-time savers want a safe, fixed-return home for emergency money, and bank FDs feel out of reach.
- **Why Jar wins.** The same mandate rail can sweep daily ₹10s into an RD or FD at partner small finance banks. Jar already works with Unity Small Finance Bank for UPI. DICGC insurance up to ₹5 lakh per depositor per bank is a regulated trust anchor Jar can explain simply.
- **Integration path.** Bank-partner FD/RD marketplace → a "safe jar" next to the gold jar → auto-split rules (e.g., 70% gold, 30% deposit).
- **Monetisation.** Distribution fee from partner banks; cross-sell lift.
- **Key risk.** Partner concentration; outsourcing and KYC obligations; keeping deposit messaging separate from gold.
- **Success metric.** Deposit balance per active saver; share of savers holding both gold and a deposit.

D. Sachet insurance on the AutoPay rail

- **User problem.** Tier-2/3 and gig workers are under-insured. Annual premiums feel large; ₹2–5 a day does not.
- **Why Jar wins.** Daily auto-debit is exactly how sachet premiums need to be collected, and Jar's plain-language style suits explaining cover.
- **Integration path.** IRDAI corporate-agent licence or broker partner → add-on toggles (personal accident, hospital cash) when a saving streak is set up.
- **Monetisation.** Commission within IRDAI expense limits.
- **Key risk.** Mis-selling or claim rejections would erode the core trust asset.
- **Success metric.** Attach rate; claim turnaround; 13-month persistency.

E. Gold-backed micro-credit (partner-led) — gated

- **User problem.** Small emergencies are funded by informal lenders, or by selling gold at a bad moment.
- **Why Jar wins.** The collateral is already vaulted and valued daily, so a small loan needs no branch visit or assayer.
- **Integration path.** Jar as lending service provider to an RBI-regulated bank or NBFC → "borrow instead of sell" offered on the sell screen.
- **Monetisation.** Sourcing fee or revenue share on disbursements.
- **Key risk.** The highest regulatory load of the five: digital-lending and gold-loan rules, plus the open question of pledging *digital* gold as collateral. Start only after the regulated rails (A, C) are live.
- **Success metric.** Loans per 1,000 active savers; 90+ day delinquency; share of borrowers who keep saving.

Deprioritised. Digital silver (same regulatory exposure, little new value); B2B payroll saving (long sales cycles).

6.3 Sequenced roadmap

Horizon	Moves	Milestone
0–6 months	B (goals → Nek) and A (SEBI-regulated gold ETF/FoF); publish a digital-gold disclosure standard	Regulated product live; first goal cohorts redeemed at Nek
6–15 months	C (insured deposits), then D (sachet insurance)	Majority of new savings on regulated or insured rails
12+ months, gated	E (gold-backed credit) via a regulated partner	Clean compliance; delinquency within plan

6.4 Sources (Section 6)

TechCrunch, "Indian fintech Jar turns profitable by helping millions save in gold", 18 Sep 2025: users, tier-2/3 share, languages, FY2024 operating revenue, Nek revenue. · Jar website (myjar.app): Brink's (vault) and Vistra (trustee) in the "Secured by" badge; ICICI Lombard named as insurer in body text, not on the "100% Insured" badge. Accessed Oct 2026. · Inc42, "India's Digital Gold Rush Gets Regulatory Reality Check", 12 Nov 2025; Groww explainer: SEBI advisory of 8 Nov 2025 and calls for standardised disclosure. · Entrackr: Jar enters UPI payments through BharatPe and Unity Small Finance Bank, 2025.

Appendix A — Assumptions log

#	Assumption	Why	Effect if wrong
A1	Order Date datetime cells have day and month swapped; text cells are dd-mm-yyyy	Only reading that makes dates monotonic in sequential Order ID	Part 2 monthly actuals would shift months
A2	Sales Target "day" component encodes the year (18 → 2018)	Months then run Apr-2018 to Mar-2019, exactly the order period	No overlap for target-vs-actual
A3	Target values are monthly <i>sales</i> (Amount) targets in ₹	Same order of magnitude as monthly category sales	Attainment metrics would not apply
A4	"Average profit per order" = profit ÷ distinct orders containing the group	Matches the brief's wording; per-line reported alongside	Magnitudes change; rankings do not
A5	Lower realised unit price on loss lines may reflect discounting (hypothesis)	No discount column exists	Product mix explains the same pattern; loss concentration still holds
A6	3 "Delhi, Madhya Pradesh" orders kept as recorded	Brief analyses the State field	Madhya Pradesh loses 3 orders; still #1
A7	Re-phased target uses same-year, company-wide seasonality, which includes Furniture	Only one year of data	MAPE gain is an upper bound; ex-Furniture sensitivity: 27.3%
A8	Q3 impact/effort scores and sizing inputs are analyst judgement, including a 20% active share of registered users and a 10% jewellery gross margin	Jar does not publish active users or unit economics	Sizes scale linearly with the active share; ranking is a starting point for discussion

Appendix B — Detail tables

B1. Sub-category performance

Category	Sub-category	Sales	Profit	Margin	Lines	Orders	Loss lines	Profit / line
Clothing	Trousers	₹30,039	₹2,847	9.5%	39	37	38.5%	₹73.00
Clothing	Stole	₹18,546	₹2,559	13.8%	192	157	27.1%	₹13.33
Clothing	Hankerchief	₹14,608	₹2,098	14.4%	198	138	21.2%	₹10.60
Clothing	T-shirt	₹7,382	₹1,500	20.3%	77	70	15.6%	₹19.48
Clothing	Shirt	₹7,555	₹1,131	15.0%	69	66	17.4%	₹16.39
Clothing	Saree	₹53,511	₹352	0.7%	210	156	46.7%	₹1.68
Clothing	Leggings	₹2,106	₹260	12.3%	53	49	30.2%	₹4.91
Clothing	Skirt	₹1,946	₹235	12.1%	64	56	28.1%	₹3.67
Clothing	Kurti	₹3,361	₹181	5.4%	47	41	29.8%	₹3.85
Electronics	Printers	₹58,252	₹5,964	10.2%	74	67	33.8%	₹80.59
Electronics	Accessories	₹21,728	₹3,559	16.4%	72	65	22.2%	₹49.43
Electronics	Phones	₹46,119	₹2,207	4.8%	83	71	49.4%	₹26.59
Electronics	Electronic Games	₹39,168	-₹1,236	-3.2%	79	73	50.6%	-₹15.65
Furniture	Bookcases	₹56,861	₹4,888	8.6%	79	72	27.8%	₹61.87
Furniture	Furnishings	₹13,484	₹844	6.3%	73	66	41.1%	₹11.56
Furniture	Chairs	₹34,222	₹577	1.7%	74	64	52.7%	₹7.80
Furniture	Tables	₹22,614	-₹4,011	-17.7%	17	16	64.7%	-₹235.94

B2. Furniture: target vs actual by month

Month	Target	Actual	Gap	Attainment	Furniture profit	Re-phased target
Apr 2018	₹10,400	₹8,121	-₹2,279	78.1%	-₹3,425	₹10,079
May 2018	₹10,500	₹6,220	-₹4,280	59.2%	-₹794	₹8,792
Jun 2018	₹10,600	₹5,532	-₹5,068	52.2%	-₹856	₹7,287
Jul 2018	₹10,800	₹3,483	-₹7,317	32.2%	-₹457	₹3,993
Aug 2018	₹10,900	₹9,538	-₹1,362	87.5%	₹443	₹9,517
Sep 2018	₹11,000	₹8,704	-₹2,296	79.1%	-₹2,468	₹8,201
Oct 2018	₹11,100	₹6,766	-₹4,334	61.0%	-₹1,316	₹9,737
Nov 2018	₹11,300	₹15,165	₹3,865	134.2%	₹3,945	₹14,810
Dec 2018	₹11,400	₹9,474	-₹1,926	83.1%	₹187	₹11,574
Jan 2019	₹11,500	₹21,257	₹9,757	184.8%	₹3,284	₹18,923
Feb 2019	₹11,600	₹16,262	₹4,662	140.2%	₹2,168	₹11,834
Mar 2019	₹11,800	₹16,659	₹4,859	141.2%	₹1,587	₹18,152

B3. All states

Rank	State	Orders	Sales	Profit	Margin	Profit / order	Sales share	Profit share
1	Madhya Pradesh	101	₹1,05,140	₹5,551	5.3%	₹54.96	24.4%	23.2%
2	Maharashtra	90	₹95,348	₹6,176	6.5%	₹68.62	22.1%	25.8%
3	Rajasthan	32	₹21,149	₹1,257	5.9%	₹39.28	4.9%	5.2%
4	Gujarat	27	₹21,058	₹465	2.2%	₹17.22	4.9%	1.9%
5	Punjab	25	₹16,786	-₹609	-3.6%	-₹24.36	3.9%	-2.5%
6	Delhi	22	₹22,531	₹2,987	13.3%	₹135.77	5.2%	12.5%
7	Uttar Pradesh	22	₹22,359	₹3,237	14.5%	₹147.14	5.2%	13.5%
8	West Bengal	22	₹14,086	₹2,500	17.7%	₹113.64	3.3%	10.4%
9	Karnataka	21	₹15,058	₹645	4.3%	₹30.71	3.5%	2.7%
10	Kerala	16	₹13,459	₹1,871	13.9%	₹116.94	3.1%	7.8%
11	Bihar	16	₹12,943	-₹321	-2.5%	-₹20.06	3.0%	-1.3%
12	Andhra Pradesh	15	₹13,256	-₹496	-3.7%	-₹33.07	3.1%	-2.1%
13	Nagaland	15	₹11,903	₹148	1.2%	₹9.87	2.8%	0.6%
14	Jammu and Kashmir	14	₹10,829	₹8	0.1%	₹0.57	2.5%	0.0%
15	Haryana	14	₹8,863	₹1,325	14.9%	₹94.64	2.1%	5.5%
16	Himachal Pradesh	14	₹8,666	₹656	7.6%	₹46.86	2.0%	2.7%
17	Goa	14	₹6,705	₹370	5.5%	₹26.43	1.6%	1.5%
18	Sikkim	12	₹5,276	₹401	7.6%	₹33.42	1.2%	1.7%
19	Tamil Nadu	8	₹6,087	-₹2,216	-36.4%	-₹277.00	1.4%	-9.3%

B4. Cities in the top-5 and loss-making states

State	City	Orders	Sales	Profit	Margin	Share of state sales	Loss lines
Andhra Pradesh	Hyderabad	15	₹13,256	-₹496	-3.7%	100.0%	26.2%
Bihar	Patna	16	₹12,943	-₹321	-2.5%	100.0%	38.7%
Gujarat	Ahmedabad	17	₹14,230	-₹880	-6.2%	67.6%	40.3%
Gujarat	Surat	10	₹6,828	₹1,345	19.7%	32.4%	8.0%
Madhya Pradesh	Indore	76	₹79,069	₹4,159	5.3%	75.2%	36.0%
Madhya Pradesh	Bhopal	22	₹23,583	₹871	3.7%	22.4%	39.4%
Madhya Pradesh	Delhi	3	₹2,488	₹521	20.9%	2.4%	0.0%
Maharashtra	Mumbai	68	₹61,867	₹1,637	2.6%	64.9%	41.1%
Maharashtra	Pune	22	₹33,481	₹4,539	13.6%	35.1%	28.9%
Punjab	Chandigarh	16	₹12,279	-₹1,153	-9.4%	73.2%	42.2%
Punjab	Amritsar	9	₹4,507	₹544	12.1%	26.8%	13.3%
Rajasthan	Udaipur	13	₹11,073	₹2,010	18.2%	52.4%	10.0%

State	City	Orders	Sales	Profit	Margin	Share of state sales	Loss lines
Rajasthan	Jaipur	19	₹10,076	-₹753	-7.5%	47.6%	27.3%
Tamil Nadu	Chennai	8	₹6,087	-₹2,216	-36.4%	100.0%	40.0%

Appendix C — Reproducibility

- **Run:** `python main.py`. This regenerates every table in `outputs/tables/`, every figure in `outputs/figures/`, `outputs/report_numbers.json`, the report Markdown and HTML in `report/`, and this PDF. Every number in this report is injected from `report_numbers.json`, and the build fails on any unresolved placeholder or broken claim. The Q1 analysis is also walked through step by step in `Q1_Sales_Analysis.ipynb`.
- **Environment:** Python 3.12.1; pandas 3.0.6; matplotlib 3.11.2; PDF rendered by headless Chromium.
- **Data:** raw Excel files in `data/` are read-only; all repairs happen in memory (Section 1.2).